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MILITARY SPECIFICATION

DISPLAY-PLOTTING BOARD GROUP OA-1768(*)/ASA-13

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide a display of ground track of an aircraft and target position data during ASW (Anti-Submarine Warfare) maneuvers

*1.2 Classification - The Display-Plotting Board Group OA-1768(*)/ASA-13 shall consist of the following items:

Item	Type Designation	Appl. Para.
* Plotting Board, Tactical Display	PT-434(*)/ASA-13	3.4.1
Control, Plotting Board, Local	C-2403(*)/ASA-13	3.4.2
Control, Plotting Board, Remote	C-2404(*)/ASA-13	3.4.3

* To include 1 roll of Film and Spool Cartridge Assembly.

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted

FSC 5841

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MIL-P-7788	Plate, Plastic, Lighting
MIL-E-7894	Electric Power, Aircraft, Characteristics of
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-D-18300	Design Data Requirements for Contracts Covering Airborne Avionic Equipment
MIL-T-18303	Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment

STANDARDS

Military

MS 25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single Phase AC, or 115/200 Volt 3-Phase AC, or Combinations

2.2 Availability of Documents

(1) When requesting applicable documents give both the title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code (CDS), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.6)

(3) When previously produced models of this equipment did not use non-repairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

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3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

* 3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 35.75 pounds.

3.2.3 Operating Life -

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 25 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.3.2 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.3.3 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3.4 Mean Operating Life Between Failures - The equipment shall have a minimum of 300 hours of mean operating life between failures. A failure shall be defined as any malfunctioning which causes the equipment to fail to meet the minimum performance requirements specified for the equipment.

3.2.4 Cabling and Connections -

3.2.4.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.4.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.4.3 Input Power Connection - Input power connections shall be in accordance with MS 25245.

3.2.5 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

* 3.2.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181, except that in the switching modes a maximum of 50 db above the limits of the specification is authorized. (See 6.8)

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3.2.7 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.8 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.9 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.10 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.11 Standby Provisions - In the STBY mode, the crab shall remain stationary but all other circuits shall be energized the same as in the RUN mode.

3.2.12 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient (up to 90% relative humidity)
Input power voltage	115 ± 1.0 V AC and 27.5 ± 0.5 V DC

3.2.13 Service Conditions - The equipment shall operate under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, with the following additions or exceptions:

3.2.13.1 Temperature (operating) - From -20 to +55°C.

3.2.13.2 Altitude - Up to 30,000 ft.

* 3.2.13.3 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve II of Specification MIL-E-5400 with respect to the resonance survey and resonance vibration. For the cycling test from 5-500 cps however, the requirements of Curve IV of Specification MIL-E-5400 shall be imposed.

3.2.13.4 Environmental - Testing for saltspray or explosion-proof capability shall not be required.

* 3.2.14 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance when provided with power from the following sources with characteristics as defined in Specification MIL-E-7894. The power required shall not exceed the amounts shown.

(1) 115 V AC 1Ø	100 VA
(2) 28 V DC	4 Amps

* 3.2.14.1 Power Supply Frequency Deviation - The equipment shall not be damaged by power supply frequencies down to 360 cps and shall automatically resume normal operation when the frequency returns within limits; however, there are no requirements on equipment performance at frequencies above or below the established operation limits.

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3.2.14.2 Ground Input - The following power lines shall be grounded externally.

(1) One side of the single-phase power source.

(2) The negative of the 27.5 V DC power source.

3.2.14.3 Primary Power Control - Primary power for the equipment shall be controlled by a switch located on Remote Plotting Board Control C-2404(*)/ASA-13.

3.3 Performance -

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerance or values setting forth acceptable variations from the performance under the standard conditions will be specified.

* 3.3.2 Warm-Up Time - The warm-up time for the equipment shall not exceed 15 minutes.

3.3.3 Design - The equipment shall continuously display aircraft present position and heading and intermittently display target position relative to present position or other stored references. An essential feature of the equipment is a "crab" which shall provide an illuminated arrow image to display present aircraft position and heading. A permanent trace visible on the plotting area to record ground track, and permanent marks used as position references, shall not be obscured by the tracing mechanism.

3.3.4 Inputs - The equipment shall accept present position data in coordinate form (N-S, E-W) and heading data, in the form of synchro signals. Target data inputs shall be in polar coordinates manually inserted. The target reference shall be selectable.

3.3.5 Outputs - The equipment shall display:

(1) Present aircraft position

(2) Ground track

(3) Heading

(4) Position references

(5) Target position

3.3.6 Scale Factors - The equipment shall operate with scale factors of one, four, and twenty nautical miles per inch. Change in scale factor shall not result in displacement of the crab.

3.3.7 Position Followup - The crab shall follow input signals up to 350 knots on all scales.

3.3.8 Heading Followup - The crab heading indicator shall follow heading signal changes up to 18 degrees-per-second.

3.3.9 Accuracy - Accuracy of the equipment shall be defined in terms of the following parameters:

(1) Position followup accuracy.

(2) Heading accuracy and followup

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(3) Target insertion accuracy and followup

(4) Centering accuracy and followup

*3.3.9.1 Accuracy Under Ambient Conditions - The following accuracies shall apply under ambient conditions, as defined in 3.2.12:

(1) Position followup accuracy - Ground track synchro inputs equivalent to one mile (180 degree synchro rotation) shall displace the crab one inch ± 1/64 inch, at a scale factor of one nautical mile per inch, in both the N-S and E-W directions, as averaged for motion in both directions. When a reference position near the center of the plotting area is approached from opposite directions and marked on the plotting film by the crab, the distance between the crab mark centers shall not exceed 1/32 inch for either N-S or E-W ground track inputs. The measurement shall be made by marking the crab position, near the center of the plotter, with the crab traveling slowly in an N-S or E-W direction to keep the follow servo in step with the simulator synchro input. After establishing a reference point on the simulator synchro corresponding to the position of the crab, continue crab travel in the same direction for approximately 1 mile past the crab mark. Then reverse the synchro direction, return the simulator synchro slowly to the reference point previously established, and mark the crab position. The distance between crab mark centers shall not exceed 1/32 inch. Measurement shall be made independently for both N-S and E-W ground tracking, with Tactical Display Plotting Board PT-434(*)/ASA-13 in either a horizontal or vertical position.

(2) Heading accuracy and followup - With a heading synchro simulator set to electrical zero, a line passing through the marking stylus position and bisecting the image of the heading arrow shall fall within ± 3 degrees of the N-S axis of the plotting surface. When the heading synchro simulator is upset 90 degrees at a minimum rate of 18 degrees per second, the heading arrow shall restabilize at the new heading within 5 seconds.

(3) Target insertion accuracy and followup - Polar coordinates shall be inserted into Remote Plotting Board Control C-2404(*)/ASA-13 at a scale factor of one, four, and twenty miles-per-inch, in both the Target Centered and Off-Course Target modes. The crab shall be positioned within an error circle having a radius of 1/16 inch to the location of the target vector. Measurement shall be made by marking the crab or target position (depending on the mode of operation) and comparing with an overlay containing the theoretical locus. During target insertion, crab slew shall be at a minimum rate of 2000 yards per second. Insertion shall be at 2000 yards per range mile. Measurements shall be made with the plotter in a vertical position.

(4) Centering accuracy and followup - At any heading, the crab shall travel to within 1/32 inch of the center within four seconds, when the centering switch is actuated.

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3.3.9.2 Accuracy Under Environmental Conditions - The following accuracies shall apply at -20°C and +55°C:

(1) Position followup accuracy - ± 1/16 inch

(2) Heading accuracy and followup - ± 5 degrees, 10 seconds

(3) Target insertion accuracy and followup - Error circle radius, 3/32 inch, 1500 yards per second; these accuracy requirements are valid for target insertions of 250 yards range or greater.

(4) Centering accuracy and followup - Centering within 1/16 inch; followup within eight seconds.

3.4 Detail Requirements -

3.4.1 Plotting Board, Tactical Display PT-434(*)/ASA-13 - The tactical display plotting board shall conform to Figure 1 and meet the following requirements:

3.4.1.1 Function - The plotting board shall accept positional data in synchro form at 180 degrees per nautical mile and shall continuously position the crab to correspond to the displacement of the aircraft, from any initial fix. The crab shall provide an illuminated arrow image, continuously oriented to correspond to aircraft heading. The crab shall be capable of marking, by means of a remotely controlled stylus, the ground track of the aircraft on a pressure-sensitive film supported directly beneath the plotting surface. The thickness of the line marked by either of the two stylii shall be between 0.004 inch and 0.010 inch wide. The crab shall also be capable of marking target or position fixes by means of cable drives, coupled through a differential to N-S and E-W position servos. Position reference shall be maintained by means of single-turn potentiometers coupled to the N-S and E-W drives. Target position shall be inserted by reference to the crab position potentiometers, to off-course target potentiometers, or to external reference potentiometers. The crab position potentiometer shall be a 2000 ohm size and the off-course target potentiometer shall be a 5000 ohm size. Both potentiometers shall be 0.1% linear and, along with the external reference potentiometer, shall be provided with 26 V AC excitation.

3.4.1.2 Form Factor and Mounting - The dimensions of the plotting board shall not exceed those given in Figure 1. The unit shall be capable of being mounted with or without vibration isolators, vertically, horizontally, or at any intermediate position. A bank of indicator lights shall be provided and shall be contained in a separate structure, capable of being mounted at the sides or bottom of the plotting board. The plotting board shall be capable of operating and being mounted as one integral unit or as two separate modules. The rear section containing the servo amplifiers, relays, and control circuits, shall be separable from the display section containing the drive and recording mechanisms. When separate mounting is employed all interconnecting cables should be suitably shielded.

3.4.1.3 Contents of Unit - The plotting board shall contain the following major assemblies to fulfill the functions described in 3.4.1.1:

(1) Bezel assembly - containing pressure-sensitive film, feed and take-up spools, and film transport controls.

(2) Drive assembly - containing multi-range gear changer assemblies, heading and marking servo, reference potentiometers, cable drive mechanism, differential, heading-oriented crab, and crab bridge structure.

(3) Electronic control assembly - containing N-S, E-W, heading, slew and null amplifiers, program function selector, and control circuits.

3.4.1.4 Weight - The weight of this unit shall not exceed 30.0 pounds.

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3.4.1.5 Electrical Connections - Electrical connections shall be provided as necessary to interconnect this unit to associated units of this equipment. The drive assembly and electronic control assembly shall interconnect by means of mating connectors. It shall be possible to relocate the latter when required by mounting conditions.

3.4.1.6 Controls - The plotting board shall contain no operating controls other than the manual film transport knobs.

3.4.1.7 Adjustments - Installation and calibration adjustments may be provided as follows:

(1) Target range gain

(2) Target range phase

3.4.1.8 Plotting Area - The bezel plotting area shall be a minimum of 9 by 9 inches. The plotting surface shall be provided with concentric range arcs having diameters of 2, 4, 6 and 8 inches. An azimuth scale shall be provided adjacent to the 8-inch range arc.

* 3.4.1.9 Writing Film - Permanent trace and position or target references shall be marked on a pressure-sensitive mylar based film. The film shall become transparent upon application of pressure by means of a stylus. The film shall be dimensionally stable, nonhygroscopic, and maintain a readable trace at temperatures from -10°C to +65°C.

3.4.1.10 Position or Target Marking - References shall be marked on the film by means of a position marking stylus displaced 1/16 inch from the tracing stylus at the center of the crab. Upon energizing the mark control circuits, internally or externally, position followup shall cease, and the marking stylus shall rotate about the tracing stylus to generate a circle. Position and heading followup shall resume upon completion of marking.

3.4.1.11 Slewing - Local slewing of the crab shall be achieved by N-S and E-W slew motors contained in the gear changer assemblies, remotely controlled from Local Plotting Board Control C-2403(*)/ASA-13. Slew rate shall be independent of position followup. The slew rate in any direction (N-S or E-W) shall not be less than one inch per second.

3.4.1.12 Centering - The N-S and E-W slew motors shall also serve to center the crab when the proper controls are energized on the Remote Plotting Board Control C-2404(*)/ASA-13.

3.4.1.13 Target Insertion - The plotting board shall provide for target insertion in any of the following selectable modes:

(1) Center reference

(2) Off-course

(3) External reference

3.4.1.13.1 In the center reference mode, insertion of target coordinates (rho, theta) shall cause the crab to be displaced from the center of the plotter at a distance and bearing corresponding to the aircraft position relative to the target at the center of the plotter. Successive target fixes shall cause displacement of the ground track, dependent on the relative velocities of the aircraft and target.

3.4.1.13.2 In the off-course mode, the plotter shall be capable of target insertion on the one (1), four (4) and twenty (20) nautical miles per inch scale. Insertion of target coordinates shall cause the crab to interrupt the ground track, mark the target position relative to the aircraft, and resume the display of track. During the insertion interval, the displacement of the aircraft shall be stored to enable resumption of correct present position display. This requirement shall not apply if target insertions are spaced less than 20 seconds apart.

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3.4.1.13.3 In the external reference mode, target insertion shall be referenced to position data stored externally rather than aircraft position.

3.4.1.14 Border Memory - The plotter shall be capable of storing position data to retain track validity when operating off any edge of the plotting area, on the one-mile scale factor, up to a distance of 0.9 miles. Off-edge operation shall be evidenced by an indicator light.

3.4.1.15 Amplifiers - All amplifiers shall be transistorized.

3.4.1.16 Lighting - No integral lighting is provided for this unit whose functions must be perceived by eye. Therefore, plotting paper with suitable contrast and a legible trace must be provided which will be visible under the red floodlighting provided in the aircraft cockpit for this purpose.

3.4.2 Local Plotting Board Control C-2403(*)/ASA-13 - The Local Plotting Board Control shall conform to Figure 2 and shall fulfill the following requirements:

3.4.2.1 Function - The local control shall provide for control of the slew, marking and trace functions for the operator at the plotting board.

* 3.4.2.2 Form Factor and Mounting - The dimensions of the Local Control shall not exceed the dimensions shown in Figure 2 and shall provide for console type mounting in accordance with Specification MIL-C-6781.

3.4.2.3 Contents of Unit - The Local Control shall contain the operating controls outlined in paragraph 3.4.2.6, together with the required relays.

3.4.2.4 Weight - The weight of this unit shall not exceed 1.5 pounds.

3.4.2.5 - Electrical Connections - Electrical connections shall be provided as necessary to interconnect this unit to associated units of this equipment.

* 3.4.2.6 Controls - The following operational controls and indicators shall be provided:

(1) The MODE switch shall control the operation of the plotter when control has been transferred by remote control MODE switch. In the STBY position, the positioning servos shall follow the ground track input synchro but the crab shall not change position. In the RUN and TRACE positions, the crab shall follow the track of the aircraft. A trace shall be marked on the film only in the TRACE position.

(2) The SLEW switch shall be a multi-position control for local slewing of the crab. The switch shall have eight positions, corresponding to cardinal and intercardinal headings, and shall be a two-speed control.

(3) The MARK pushbutton switch, when actuated, shall cause position references to be marked on the film. The MARK control shall be paralleled with an external circuit, for automatic insertions of position references.

(4) The TARGET compliance switch shall be a combined pushbutton and indicator. When target data is inserted from the remote control, the indicator shall be energized simultaneously with a TARGET indicator on the plotting board. Acknowledgment of data transmission shall be accomplished by actuating the switch, to extinguish both indicators. Corrected or sequential target data shall again cause the indicators to be energized, thus enabling the local operator to correlate the latest available data as it appears on the plotter. When pressed, the TARGET compliance switch shall cause the C-2404(*)/ASA-13 INSERT lamp to light thus providing "signal back" capability.

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* 3.4.2.7 Lighting - Lighting for this unit shall be in accordance with Specification MIL-P-7788.

3.4.3 Remote Plotting Board Control C-2404(*)/ASA-13 - The Remote Plotting Board Control shall conform to Figure 3 and shall fulfill the following requirements:

3.4.3.1 Function - The Remote Plotting Board Control shall provide for control of the plotting board operating mode and function, and target data insertion from a remote position. The unit shall also control the position of a range strobe and a time-shared azimuth cursor, in an associated radar indicator, for deriving target data relative to the aircraft.

3.4.3.2 Form Factor and Mounting - The dimensions of this unit shall not exceed the dimensions shown in Figure 3. The Remote Plotting Board Control shall provide for console mounting in accordance with Specification MIL-C-6781.

3.4.3.3 Contents of Unit - The Remote Plotting Board Control shall contain the operating controls outlined in paragraph 3.4.3.6, together with a target azimuth synchro and resolver, strobe and target range potentiometers, and relays.

3.4.3.4 Weight - The weight of this unit shall not exceed 4.25 pounds.

3.4.3.5 Electrical Connections - Electrical connections shall be provided as necessary to interconnect this unit to other associated units of this equipment.

(1) Cursor Azimuth Output - The Remote Control shall include provision for transmitting azimuth information, corresponding to the setting of the bearing control, to the associated radar indicator. This information shall be in the form of synchro voltages from a Size 11 synchro, the rotor excitation for which is supplied by the auxiliary equipment. This information shall be available at the following receptacle pins:

Receptacle 4J1

PIN W	Azimuth rotor H
X	Azimuth rotor C
R	Azimuth stator X
S	Azimuth stator Z
P	Azimuth stator Y

These stator and rotor connections apply for clockwise (increasing azimuth) rotation of the bearing pointer on the Remote Control and are standardized in accordance with the ARINC method. The synchro output shall be factory adjusted so that electrical zero corresponds to North within ± 2 degrees.

(2) Strobe Range Output - The Remote Control shall provide an output voltage for controlling the displacement of a strobe on a time-shared cursor in the associated radar indicator. The output voltage shall be a function of the range control setting. The information shall be available at the following receptacle pins. Excitation voltage for the strobe potentiometer shall be supplied externally.

Receptacle 4J1

PIN Y	Strobe output
Z	Strobe excitation
a	Strobe excitation

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* 3.4.3.6 Controls - The following controls and indicators shall be provided:

(1) The MODE switch shall be a four-position switch. In the OFF position, no input power (except for panel lights and RFI filters) shall be consumed by the equipment. In the REM position, control shall be transferred to the local control and the REMOTE indicator light on the plotting board shall be extinguished. The STBY and RUN positions shall have the same functions as the corresponding switch positions in the local control. Local slew at the local control shall be ineffective unless the MODE switch is in the REM position.

(2) The FUNCTION switch shall be a four-position, rotary switch associated with a combined INSERT pushbutton switch and indicator in its operation. With the switch at A/C CTR, actuation of the INSERT pushbutton shall cause the crab to slew to the center of the plotting board. At TGT CTR target data set in by means of the RANGE and BEARING controls shall be referenced to the center of the plotting board when the INSERT switch is actuated. In the OCT/PP position, off-course target data shall be referenced to aircraft present position. When the INSERT switch is depressed, the crab shall slew to the position determined by the target rho-theta coordinates, mark the target position and resume ground track, without losing present position. In the OCT/RR position, off-course target data shall be referenced to an external, stored position reference, in rectangular coordinates, as represented by voltages on potentiometers.

(3) The INSERT pushbutton-indicator shall cause target data set into the remote control to displace the crab in a manner dependent on the setting of the FUNCTION switch. Insertion of target data shall cause the INSERT pushbutton-indicator to light. Actuation of the acknowledgment TARGET switch on the local control shall extinguish the indicator.

(4) The target BEARING control shall simultaneously position the time-shared cursor in azimuth, in the associated radar indicator, and provide voltages corresponding to the sine and cosine components of bearing to a dual range potentiometer.

(5) The target RANGE control shall control the displacement of a strobe on the time-shared cursor in the associated radar indicator. The control shall also set target range on a Range Counter which shall indicate ranges up to 36,000 yards maximum and 360 miles maximum. The target RANGE control shall also automatically select the proper scale factor to accommodate the range set in when the FUNCTION switch is in the TGT/CTR position, for the one mile and four mile scale factors and shall inhibit inserts greater than 80 miles in TGT/CTR for the 20 mile-per-inch scale factor.

(6) The SCALE pushbutton-indicator switches shall select the operating range of the plotting board. The 20 mile scale factor shall be automatically selected when the C-2404(*)/ASA-13 MODE switch is initially switched from OFF. Three switches corresponding to 1, 4, and 20 nautical miles per inch shall be provided and operation shall be indicated by the integral light contained in each switch. Scale factors shall be based on input synchro constants of 180° per nautical mile.

(7) The MARK pushbutton switch shall cause position references to be marked on the film.

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3.4.3.7 Adjustments - Installation and calibration adjustments shall be provided as follows:

Target Range - N-S, one, four, and twenty mile scales
E-W, one, four, and twenty mile scales

NOTE: One and twenty mile target range adjustments may be accomplished on the same control.

* 3.4.3.8 Lighting - Lighting for this unit shall be in accordance with Specification MIL-P-7788.

3.4.4 Associated Equipment - The equipment shall be capable of operating with the following equipments not supplied as part of the OA-1768(*)/ASA-13.

Item	Type Designation	Function
Compass System	G-2 N-1 GMC-2, MF-1	Supply heading data.
Computer Group, Navigational	AN/ASA-13 AN/ASA-13A AN/ASA-24	Supply position data in coordinate form
Dead Reckoning Tracer Test Set	TS-1160/U	Supply simulated heading and position data for testing
Computer Group, Target Position	AN/ASA-31(XN-1)	Auxiliary computer group
Data Display Group	AN/ASA-25 AN/ASA-16	Accept, store, generate, and distribute display data

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment. It may not be resubmitted as the production sample model.

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4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction and performance and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when required by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - After completion of the production sample tests the equipment shall be returned to the contractor for any necessary rework. The equipment shall then be resubmitted for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

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4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 8 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±2g

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

(3) Manufacturing reliability test specified in 4.4.1.3, except that the test duration shall be 120 hours (with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.)

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4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity. for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5. Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. When a production sample equipment is submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. Voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

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4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Use - The equipment shall be used in military aircraft as an airborne ASW navigational and localization device. The display, in the form of an illuminated arrow (crab), shall be continuously oriented to correspond to aircraft heading and ground track. The plotting board shall indicate as well as permanently record: the position of the aircraft from any selected origin, the position of a target relative to a remote reference point or the present position of the aircraft, and the aircraft position relative to a target.

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6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

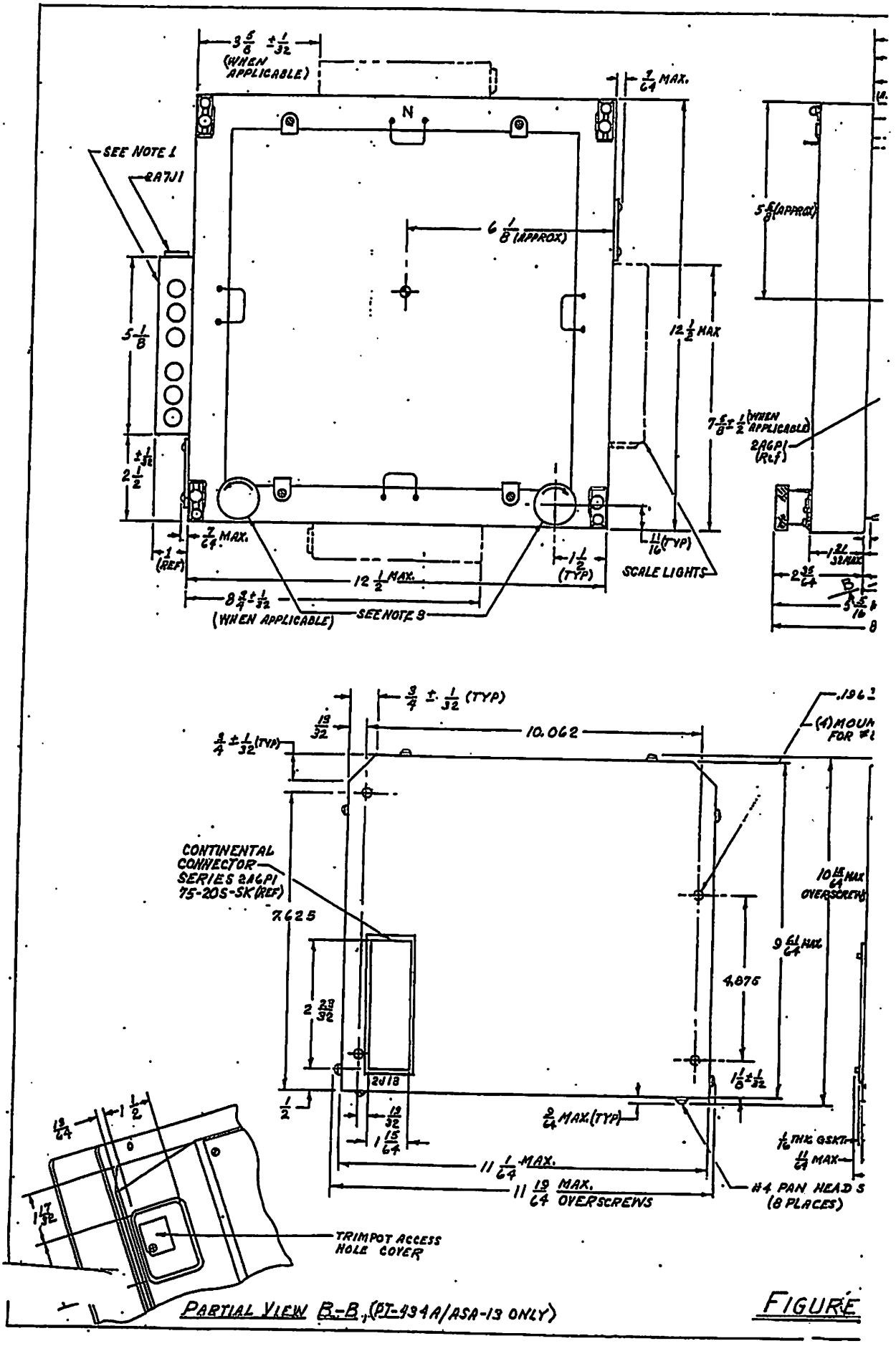
6.6 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.7 An asterisk preceding a paragraph number denotes a change in that paragraph from the basic specification.

6.8 The first procurement lot of OA-1768/ASA-13 failed to meet Specification MIL-I-6181. Filters were added which have brought the later model OA-1768A/ASA-13 to within 50 db of the limits of Specification MIL-I-6181. Tests by the aircraft manufacturers have shown no RFI problems with this deviation in the installation in P2V, P5M, S2F or P3V aircraft. There are approximately 1269 OA-1768(*)/ASA-13's in the fleet and the fleet finds the equipments to be acceptable insofar as the RFI deviation is concerned.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

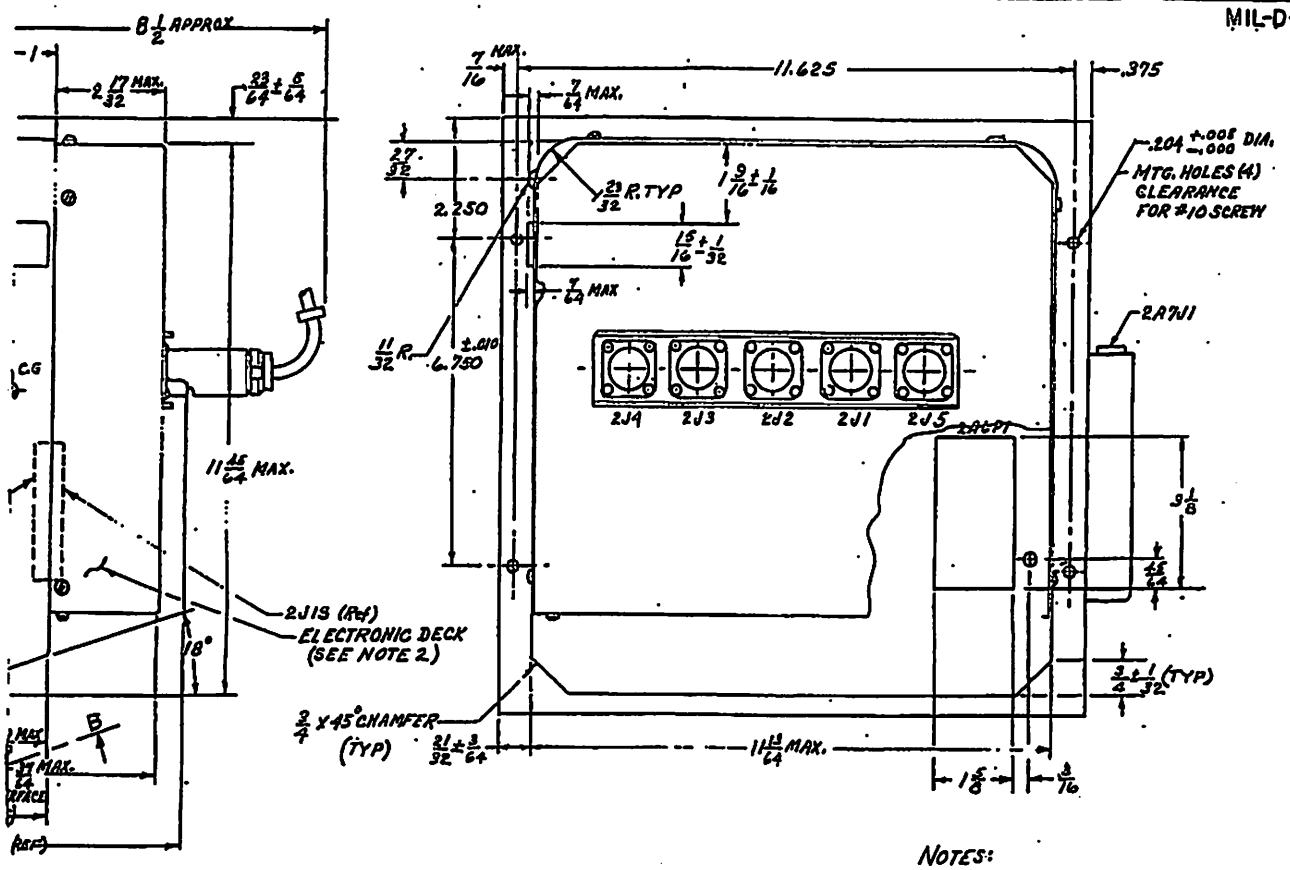
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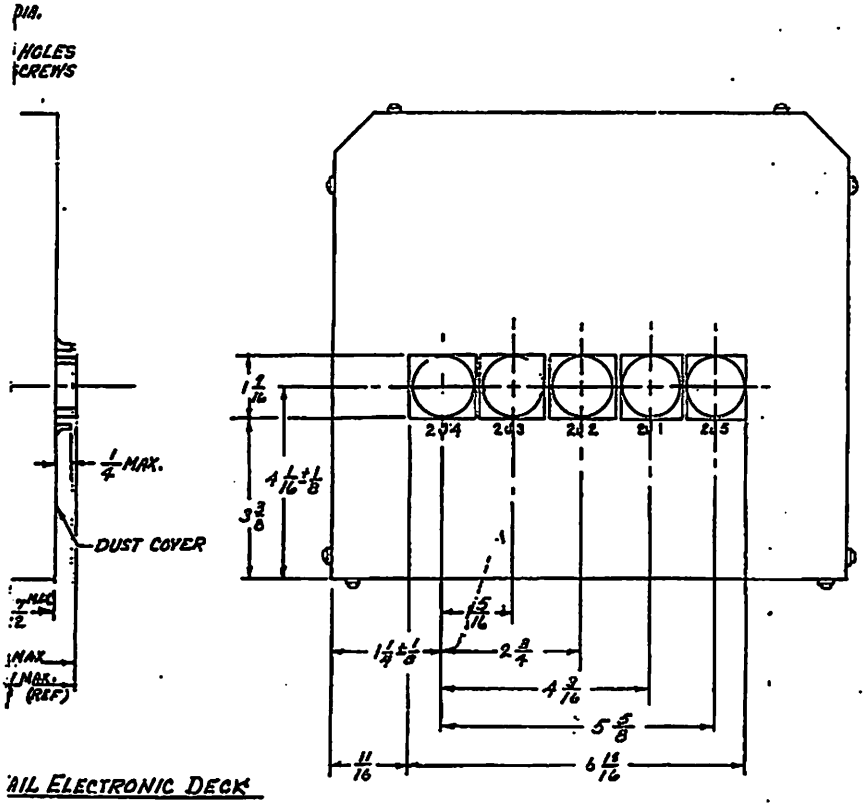


PARTIAL VIEW B-B, (PT-934A/ASA-13 ONLY)

FIGURE

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AIL ELECTRONIC DECK

NOTES:

1. THIS ASSEMBLY IS SEPARABLE AND MAY BE MOUNTED IN THE POSITION INDICATED OR IN ALTERNATE POSITION AS SHOWN IN PHANTOM.
2. THIS ASSEMBLY MAY BE MOUNTED REMOTELY FROM THE MAIN DISPLAY. MOUNTING DIMENSIONS FOR THIS UNIT ARE SHOWN IN "DETAIL ELECTRONIC DECK".
3. FILM TAKE-UP AND REVERSE KNOBS.
4. APPROXIMATE LOCATION OF MOUNTING HOLES.
5. 2J4 IS CONNECTED TO 2A7J1 BY AN EXTERNAL CABLE.
6. UNIT WEIGHT 30.0 LBS.
7. HEAT DISSIPATION:

ELECTRONIC DECK	21 WATTS.
BASE ASSEMBLY	48 WATTS.
SCALE LIGHTS	6 WATTS

CONNECTOR NO.	TYPE OR APPROVED EQUIV.
2J13	CONTINENTAL 75-20SS-K6D
2A6P1	CONTINENTAL 75-20PS-6D
2J5	AMPHENOL 165-27
2J1	AMPHENOL 165-28W
2J2	AMPHENOL 165-20
2J4	AMPHENOL 165-12
2J3	AMPHENOL 165-28X
2A7J1	CONTINENTAL MM-3MLR OR GM-3MLR

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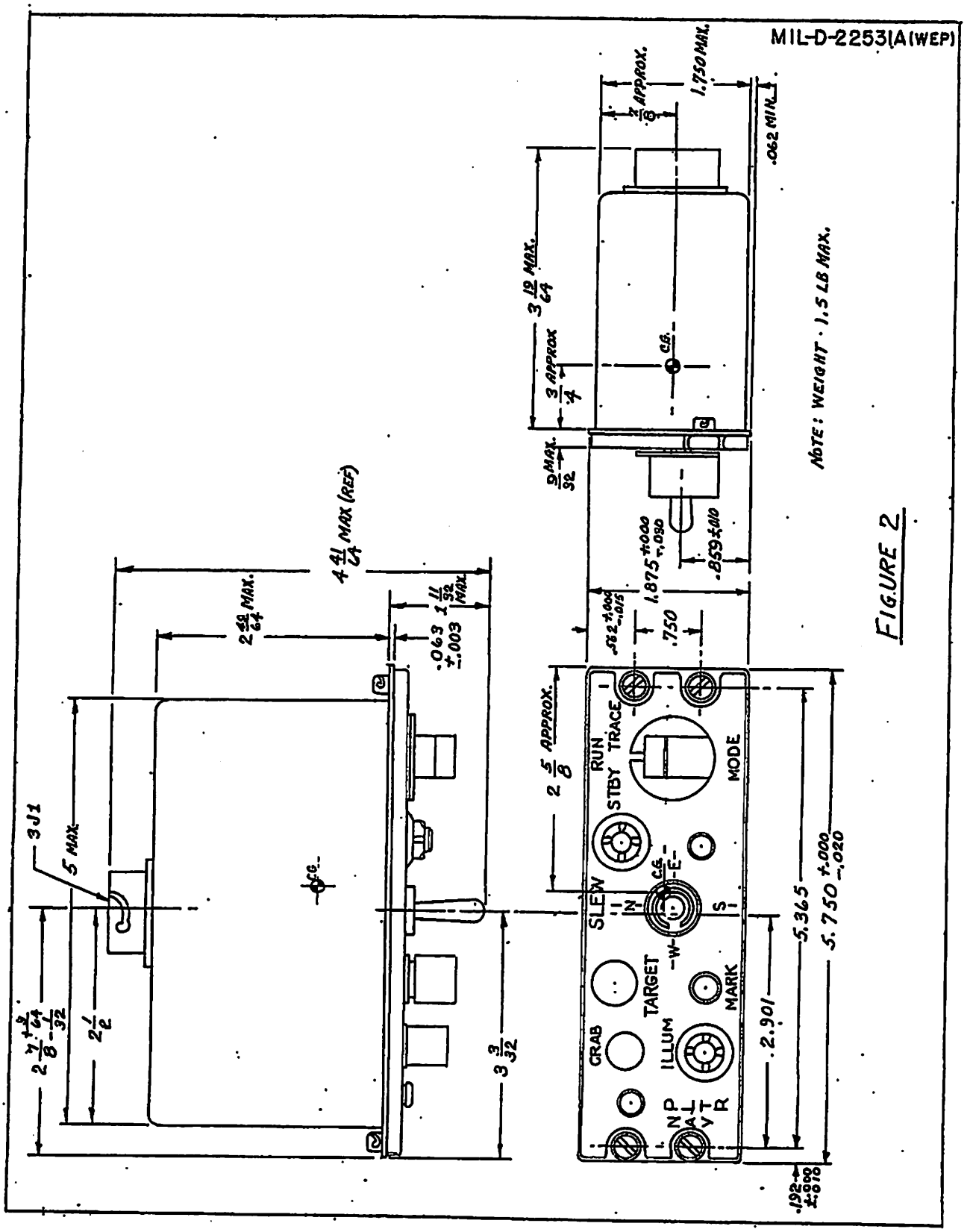


FIGURE 2

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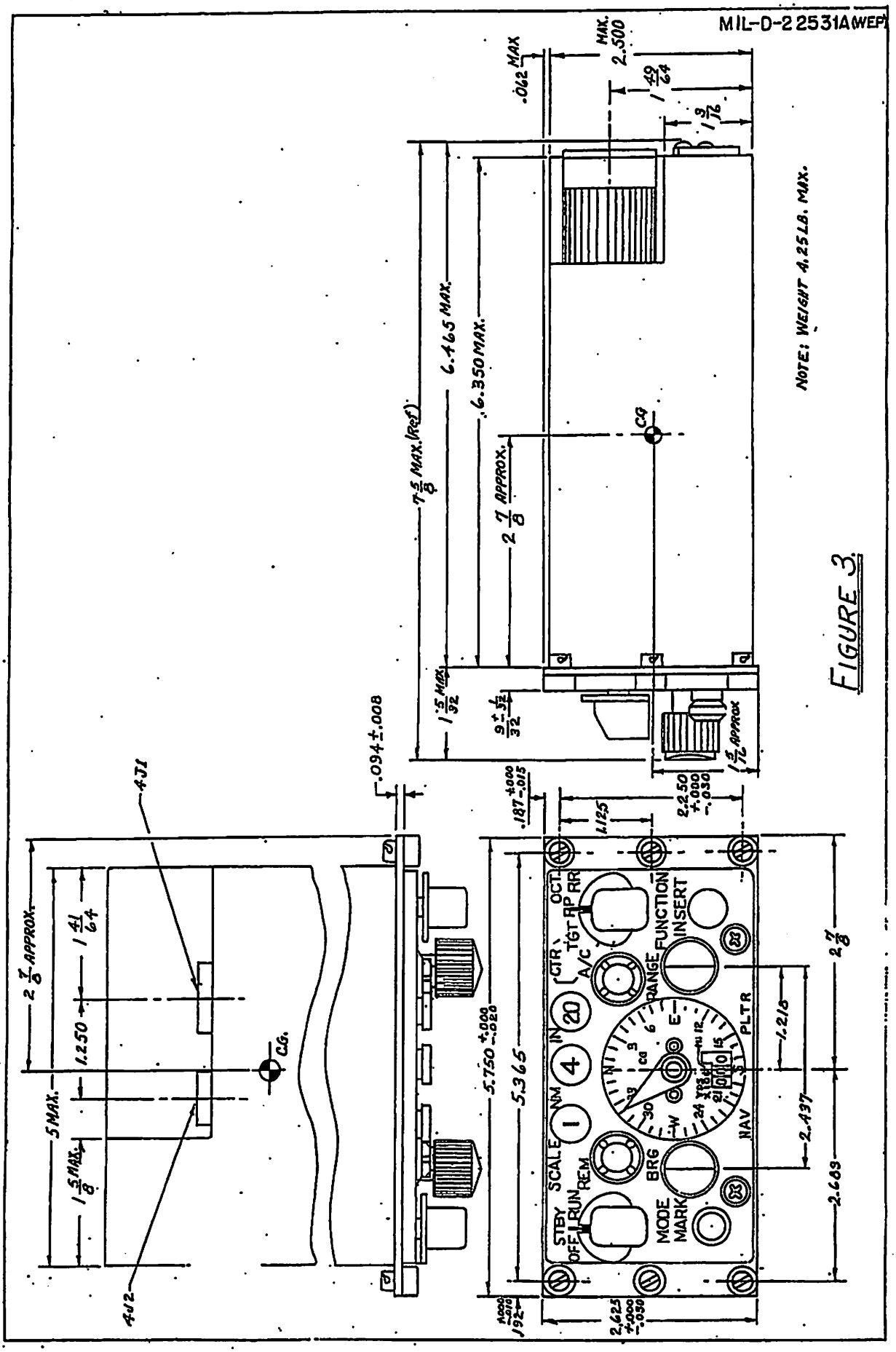


FIGURE 3.

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